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GATE ECE · COMMUNICATIONS · CHAPTER-WISE PYQS

Noise In Digital Communication

GATE ECE Communications · Previous Year Questions with Answer Key

45	26	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 2	40 questions
Answer key	at the end

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Marks 1 5 questions

Q1. GATE ECE 2017 Set 2

A sinusoidal message signal is converted to a PCM signal using a uniform quantizer. The required signal to quantization noise ratio (SQNR) at the output of the quantizer is 40dB. The minimum number of bits per sample needed to achieve the desired SQNR is _____ .

Open solution

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Q2. GATE ECE 2016 Set 3

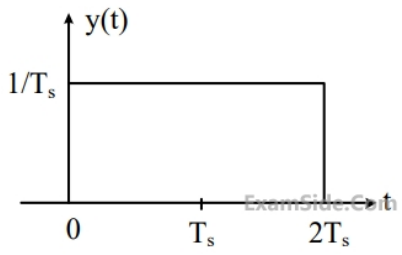
Open solution

A binary baseband digital communication system employs the signal

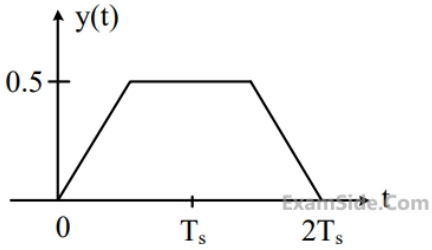
$$p(t) = \begin{cases} \frac{1}{\sqrt{T_s}}, & 0 \le t \le T_s \\ 0, & otherwise \end{cases},$$

for transmission of bits. The graphical representation of the matched filter output y(t) for this signal will be

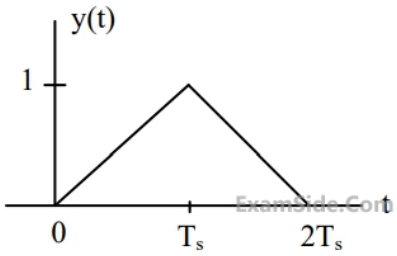
(A)



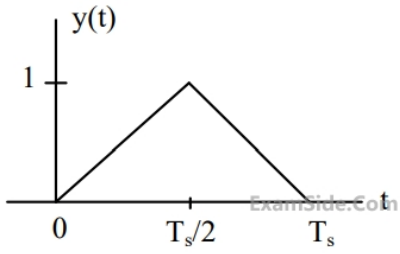
(B)



(C)



(D)



Q3. GATE ECE 2015 Set 2

Open solution

A sinusoidal signal of amplitude A is quantized by a uniform quantizer. Assume that the signal utilizes all the representation levels of the quantizer. If the signal to quantization noise ratio is 31.8 dB, the number of levels in the quantizer is _____ .

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Q4. GATE ECE 2014 Set 1

The capacity of a Binary Symmetric Channel (BSC) with cross - over probability 0.5 is _____ .

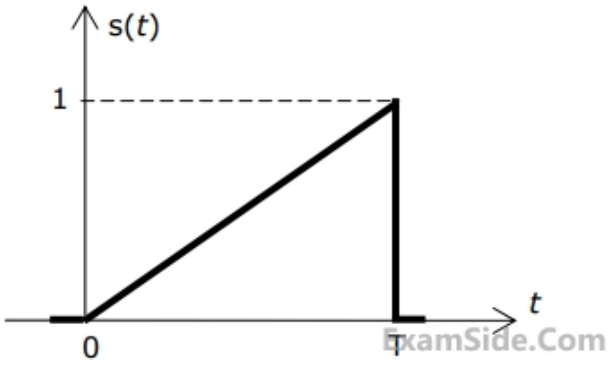
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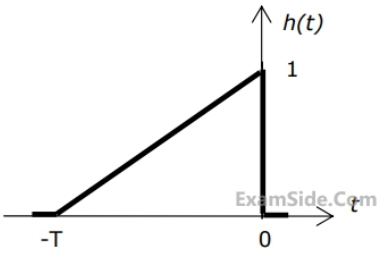
Q5. GATE ECE 2010

Consider the pulse shape s(t) as shown. The impulse response h(t) of the filter matched to this pulse is

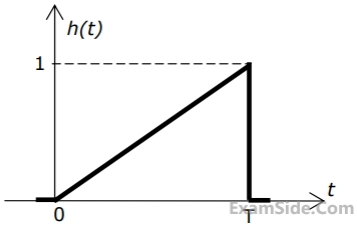
Open solution



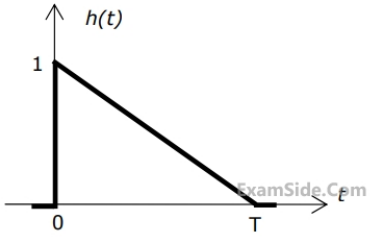
(A)



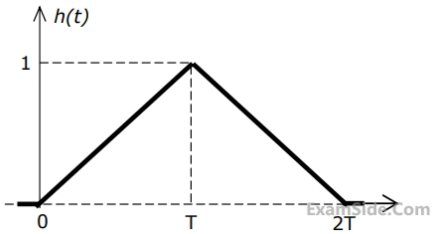
(B)



(C)



(D)



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Marks 2 40 questions

Q6. GATE ECE 2026

A QPSK modulated signal from an additive white Gaussian noise (AWGN) channel is received with an $E_b/N_o = 8.4\,\text{dB}$ at the input of a coherent QPSK demodulator. A maximum-likelihood reception method is used in the demodulator.

Assume the complimentary error function

$$\text{erfc}(u) \cong \left[\frac{1}{(u\sqrt{\pi})}\right] \exp\left(-u^2\right)$$

Which is the nearest bit error rate (BER) at the output of the demodulator?

(A) 10^{-3} (B) 10^{-4} (C) 10^{-5} (D) 10^{-6}

Open solution

Q7. GATE ECE 2025

A source transmits symbol S that takes values uniformly at random from the set $\{-2, 0, 2\}$. The receiver obtains $Y = S + N$, where N is a zero-mean Gaussian random variable independent of S. The receiver uses the maximum likelihood decoder to estimate the transmitted symbol S.

Suppose the probability of symbol estimation error P_e is expressed as follows:

$P_e = \alpha P(N > 1)$,

where $P(N > 1)$ denotes the probability that N exceeds 1 .

What is the value of α ?

(A) $\frac{1}{3}$ (B) 1 (C) $\frac{2}{3}$ (D) $\frac{4}{3}$

Open solution

Q8. GATE ECE 2021

Consider a polar non-return to Zero (NRZ) waveform, using +2 V and -2 V for representing binary ' 1 ' and ' 0 ' respectively, is transmitted in the presence of additive zero-mean white Gaussian noise with variance $0.4\,\text{V}^2$. If the a priori probability of transmission of a binary ' 1 ' is 0.4 , the optimum threshold voltage for a maximum a posteriori (MAP) receiver (rounded off to two decimal places) is ____ V.

Open solution

Q9. GATE ECE 2016 Set 1

A digital communication system uses a repetition code for channel encoding/decoding. During transmission, each bit is repeated three times (0 is transmitted as 000, and 1 is transmitted as 111). It is assumed that the source puts out symbols independently and with equal probability. The decoder operates as follows: In a block of three received bits, if the number of zeros exceeds the number of ones, the decoder decides in favor of a 0, and if the number of ones exceeds the number of zeros, the decoder decides in favor of a 1, Assuming a binary symmetric channel with crossover probability p = 0.1, the average probability of error is _____

Open solution

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Q10. GATE ECE 2016 Set 1

Open solution

An analog pulse s(t) is transmitted over an additive white Gaussian noise (AWGN) channel. The received signal is r(t) = s(t) + n(t), where n(t) is additive white Gaussian noise with power spectral density $\frac{N_0}{2}$. The received signal is passed through a filter with impulse response h(t). Let E_s and E_n denote the energies of the pulse s(t) and the filter h(t), respectively. When the signal-to-noise ratio (SNR) is maximized at the output of the filter $(SNR_{\max})$, which of the following holds?

(A) $E_s = E_h;\ SNR_{\max} = \frac{2E_s}{N_0}$

(B) $E_s = E_h;\ SNR_{\max} = \frac{E_s}{2N_0}$

(C) $E_s > E_h;\ SNR_{\max} > \frac{2E_s}{N_0}$

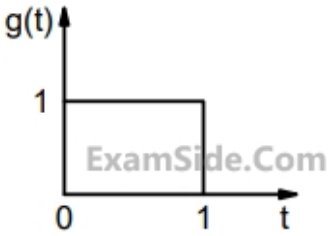
(D) $E_s < E_h;\ SNR_{\max} = \frac{2E_h}{N_0}$

Q11. GATE ECE 2015 Set 2

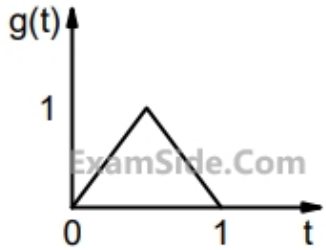
Open solution

Consider a binary, digital communication system which uses pulses g (t) and − g (t)for transmitting bits over an AWGN channel. If the receiver uses a matched filter, which one of the following pulses will give the minimum probability of bit error?

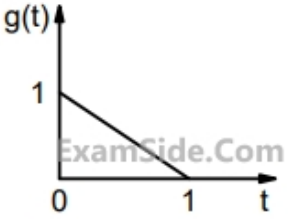
(A)



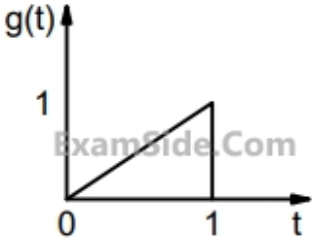
(B)



(C)



(D)



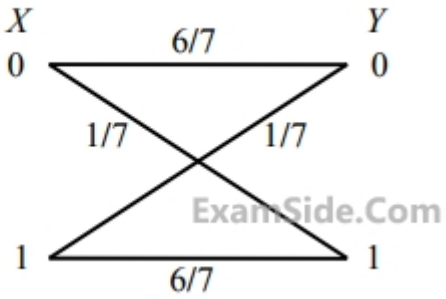
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Q12. GATE ECE 2015 Set 1

The input X to the Binary Symmetric Channel (BSC) shown in the figure is '1' with probability 0.8. The cross-over probability is 1/7. If the received bit Y = 0, the conditional probability that '1' was transmitted is _____.

Open solution

$P[X=0]=0.2$



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Q13. GATE ECE 2015 Set 1

A source emits bit 0 with probability $\frac{1}{3}$ and bit 1 with probability $\frac{2}{3}$. The emitted bits are communicated to the receiver. The receiver decides for either 0 or 1 based on the received value R. It is given that the conditional density functions of R are as

$$f_{R|0}(r)=\begin{cases}\frac{1}{4}, & -3 \le r \le 1,\\ 0 & otherwise,\end{cases} \text{ and}$$

$$f_{R|1}(r)=\begin{cases}\frac{1}{6}, & -1 \le r \le 5,\\ 0 & otherwise.\end{cases}$$

The minimum decision error probability is

(A) 0 (B) 1/12 (C) 1/9 (D) 1/6

Open solution

Q14. GATE ECE 2014 Set 4

Consider a communication scheme where the binary valued signal X satisfies P{X = + 1} = 0.75 and P {X = - 1} = 0.25. The received signal Y = X + Z, where Z is a Gaussian random variable with zero mean and variance σ^2. The received signal Y is fed to the threshold detector. The output of the threshold detector $\hat{X}$ is:

$$\hat{X}:\begin{cases}+1, & Y > \tau\\ -1, & Y \le \tau.\end{cases}$$

To achieve a minimum probability of error $P\{\hat{X} \neq X\}$, the threshold τ should be

(A) strictly positive

(B) zero

(C) strictly negative

(D) strictly positive, zero, or strictly negative depending on the nonzero value of σ^2

Open solution

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Q15. GATE ECE 2014 Set 4

Consider a discrete-time channel Y = X + Z, where the additive noise Z is signal- dependent. In particular, given the trasmitted symbol $X \in \{-a, +a\}$ at any instant, the noise sample Z is chosen independently from a Gaussian distribution with mean βX and unit variance. Assume a threshold detector with zero threshold at the receiver. When β = 0 the BER was found to be $Q(a) = 1 \times 10^{-8}$.

$\left(Q(v) = \frac{1}{\sqrt{2\pi}} \int_{v}^{\infty} e^{-u^2/2}$ du, and for v > 1,

use $Q(v) \approx e^{-v^2/2}$

When $\beta = -0.3$, the BER is closed to

(A) 10^{-7} (B) 10^{-6} (C) 10^{-4} (D) 10^{-2}

Open solution

Q16. GATE ECE 2014 Set 2

Coherent orthogonal binary FSK modulation is used to transmit two equiprobable symbol waveforms $s_1(t) = \alpha \cos 2\pi f_1 t$ and $s_2(t) = \alpha \cos 2\pi f_2 t$, where $\alpha = 4\ mV$. Assume an AWGN channel with two-sided noise power spectral density $\frac{N_0}{2} = 0.5 \times 10^{-12}$ W/Hz. Using an optimal receiver and the relation $Q(v) = \frac{1}{\sqrt{2\pi}} \int_{v}^{\infty} e^{-u^2/2}\, du$, the bit error probability for a data rate of 500 kbps is

(A) Q (2) (B) $Q\left(2\sqrt{2}\right)$ (C) Q (4) (D) $Q\left(4\sqrt{2}\right)$

Open solution

Q17. GATE ECE 2013

Let U and V be two independent zero mean Gaussian random variables of variances $\frac{1}{4}$ and $\frac{1}{9}$ respectively. The probability $P(3V \geq 2U)$ is

(A) 4/9 (B) 1/2 (C) 2/3 (D) 5/9

Open solution

Q18. GATE ECE 2012

A BPSK scheme operating over an AWGN channel with noise power spectral density of $N_0/2$, uses equi-probable signals $s_1(t) = \sqrt{\frac{2E}{T}} \sin(\omega_c t)$

and $s_2(t) = -\sqrt{\frac{2E}{T}} \sin(\omega_c t)$

over the symbol interval, $(0, T)$. If the local oscillator in a coherent receiver is ahead in phase by 45^0 with respect to the received signal, the probability of error in the resulting system is

(A) $Q\left(\sqrt{\frac{2E}{N_0}}\right)$ (B) $Q\left(\sqrt{\frac{E}{N_0}}\right)$

(C) $Q\left(\sqrt{\frac{E}{2N_0}}\right)$ (D) $Q\left(\sqrt{\frac{E}{4N_0}}\right)$

Open solution

Q19. GATE ECE 2012

A binary symmetric channel (BSC) has a transition probability of 1/8. If the binary transmit symbol X is such that P(X =0) = 9/10, then the probability of error for an optimum receiver will be

(A) 7/80 (B) 63/80 (C) 9/10 (D) 1/10

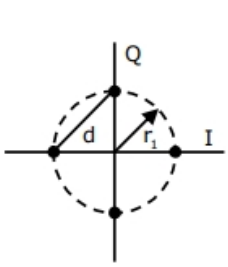
Open solution

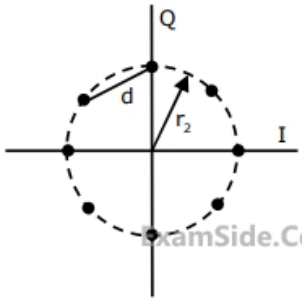
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Q20. GATE ECE 2011

Open solution

A four phase and an eight-phase signal constellation are shown in the figure below.





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For the constraint that the minimum distance between pairs of signal points be d for both constellations, the radii r_1, and r^2 of the circles are

(A) $r_1 = 0.707d, r_2 = 2.782d$

(B) $r_1 = 0.707d, r_2 = 1.932d$

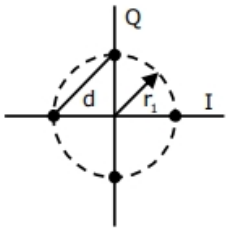
(C) $r_1 = 0.707d, r_2 = 1.545d$

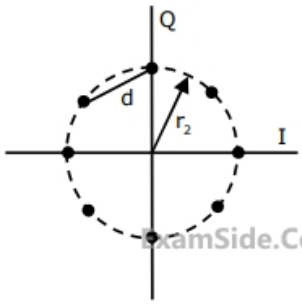
(D) $r_1 = 0.707d, r_2 = 1.307d$

Q21. GATE ECE 2011

Open solution

A four phase and an eight-phase signal constellation are shown in the figure below.





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Assuming high SNR and that all signals are equally probable, the additional average transmitted signal energy required by the 8-PSK signal to achieve the same error probability as the 4-PSK signal is

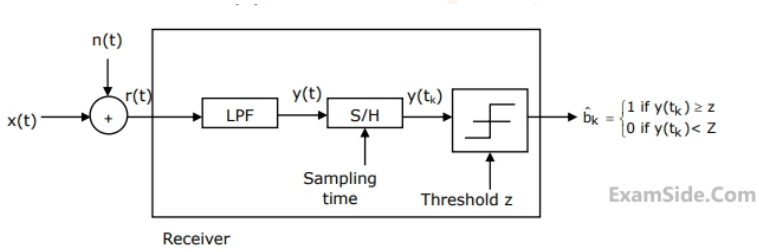
(A) 11.90 dB (B) 8.73 dB (C) 6.79 dB (D) 5.33 dB

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Q22. GATE ECE 2010

Open solution

Consider a base band binary PAM receiver shown below. The additive channel noise $n(t)$ is white with power spectral density $S_N(f) = N_0/2 = 10^{-20}\ W/Hz$. The low-pass filter is ideal with unity gain and cut -off frequency $1MHz$. Let Y_k represent the random variable $y(t_k)$.
$Y_k = N_k$ if transmitted bit $b_k = 0$
$Y_k = a + N_k$ if transmitted bit $b_k = 1$
Where $b_k = 0$ represents the noise sample value. The noise sample has a probability density function, $P_{N_k}(n) \quad = 0.5\alpha e^{-\alpha|n|}$ (This has mean zero and variance $2/\alpha^2$). Assume transmitted bits to be equiprobable and threshold z is set to $a/2 = 10^{-6}V$.



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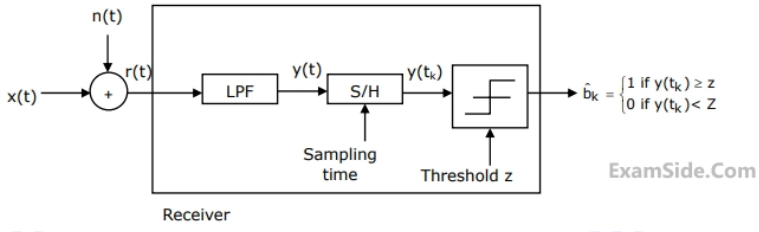
The value of the parameter α(in V^{-1}) is

(A) 10^{10} (B) 10^{7} (C) 1.414×10^{-10} (D) 2×10^{-20}

Q23. GATE ECE 2010

Open solution

Consider a base band binary PAM receiver shown below. The additive channel noise $n(t)$ is white with power spectral density $S_N(f) = N_0/2 = 10^{-20}\ W/Hz$. The low-pass filter is ideal with unity gain and cut -off frequency $1MHz$. Let Y_k represent the random variable $y(t_k)$.
$Y_k = N_k$ if transmitted bit $b_k = 0$
$Y_k = a + N_k$ if transmitted bit $b_k = 1$
Where $b_k = 0$ represents the noise sample value. The noise sample has a probability density function, $P_{N_k}(n) \quad = 0.5\alpha e^{-\alpha|n|}$ (This has mean zero and variance $2/\alpha^2$). Assume transmitted bits to be equiprobable and threshold z is set to $a/2 = 10^{-6}V$.



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The probability of bit error is

(A) $0.5 \times e^{-3.5}$ (B) $0.5 \times e^{-5}$ (C) $0.5 \times e^{-7}$ (D) $0.5 \times e^{-10}$

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Q24. GATE ECE 2009

The amplitude of random signal is uniformly distributed between -5V and 5V
If the signal to quantization noise ratio required in uniformly quantizing the signals is 43.5 dB, the step size of the quantization is approximately

Open solution

(A) 0.0333 V (B) 0.05 V (C) 0.0667 V (D) 0.10 V

Q25. GATE ECE 2009

The amplitude of random signal is uniformly distributed between -5V and 5V
If the positive values of the signal are uniformly quantized with a step size of 0.05 V, and the negative values are uniformly quantized with a step size of 0.1V, the resulting signal to quantization noise ratio is approximately

Open solution

(A) 46 dB (B) 43.8 dB (C) 42 dB (D) 40 dB

Q26. GATE ECE 2008

Consider a Binary Symmetric Channel (BSC) with probability of error being 'p'. To transit a bit, say 1, we transmit a sequence of three 1s. The receiver will interpret the received sequence to represent 1 if at least two bits are 1. The probability that the transmitted bit will be received in error is

Open solution

(A) $p^3 + 3p^2(1-p)$ (B) p^3
(C) $(1-p)^3$ (D) $p^3 + p^2(1-p)$

Q27. GATE ECE 2007

During transmission over a certain binary communication channel, bit errors occurs independently with probability p. The probability of at most one bit in error in a block of n bits is given by

Open solution

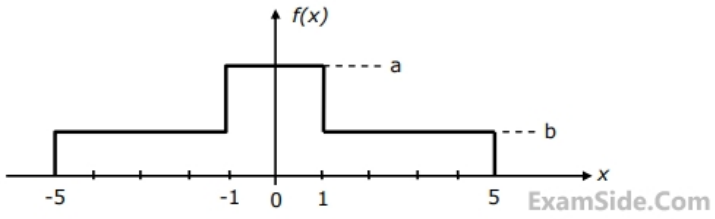
(A) p^n (B) $1-p^n$
(C) $np(1-p)^{n-1} + (1-p)^n$ (D) $1-(1-p)^n$

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Q28. GATE ECE 2007

Open solution

An input to a 6-level quantizer has the probability density function f(X) as shown in the figure. Decision boundaries of the quantizer are chosen so as to maximize the entropy of the quantizer output. It is given that 3 consecutive decision boundaries are '-1', '0' and '1'.



The values of a and b are

(A) a = 1/6 and b = 1/12

(B) a = 1/5 and b = 3/40

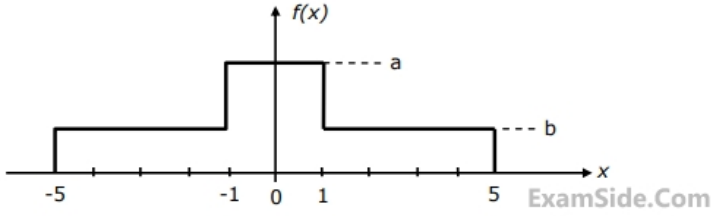
(C) a = 1/4 and b = 1/16

(D) a = 1/3 and b = 1/24

Q29. GATE ECE 2007

Open solution

An input to a 6-level quantizer has the probability density function f(X) as shown in the figure. Decision boundaries of the quantizer are chosen so as to maximize the entropy of the quantizer output. It is given that 3 consecutive decision boundaries are '-1', '0' and '1'.



Assuming that the reconstruction levels of the quantizer are the mid-points of the decision boundaries, the ratio of signal power to quantization noise power is

(A) $\frac{152}{9}$

(B) $\frac{64}{3}$

(C) $\frac{76}{3}$

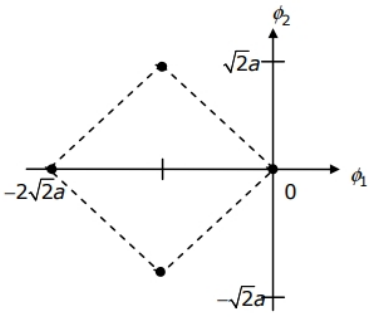
(D) 28

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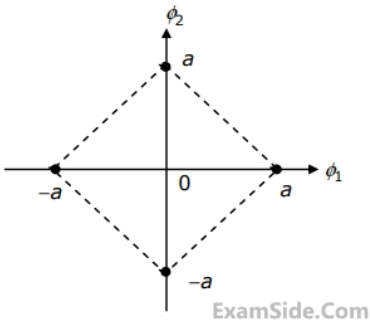
Q30. GATE ECE 2007

Open solution

Two 4-ray signal constellations are shown. It is given that ϕ_1 and ϕ_2 constitute an orthonormal basis for the two constellations. Assume that the four symbols in both the constellations are equiprobable. Let $\frac{N_0}{2}$ denote the power spectral density of white Gaussian noise.



Constellation 1



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Constellation 2

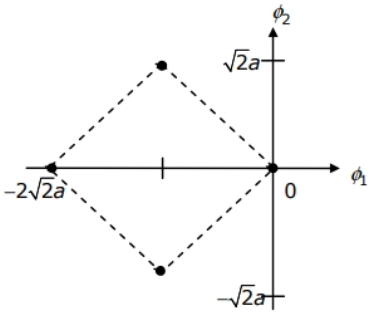
The ratio of the average energy of Constellation 1 to the average energy of Constellation 2 is

(A) $4a^2$ (B) 4 (C) 2 (D) 8

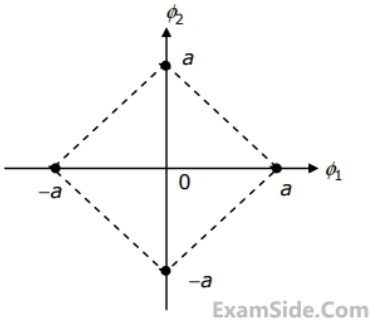
Q31. GATE ECE 2007

Open solution

Two 4-ray signal constellations are shown. It is given that ϕ_1 and ϕ_2 constitute an orthonormal basis for the two constellations. Assume that the four symbols in both the constellations are equiprobable. Let $\frac{N_0}{2}$ denote the power spectral density of white Gaussian noise.



Constellation 1



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Constellation 2

If these constellations are used for digital communications over an AWGN channel, then which of the following statements is true?

(A) Probability of symbol error for Constellation 1 is lower

(B) Probability of symbol error for Constellation 1 is higher

(C) Probability of symbol error is equal for both the constellations

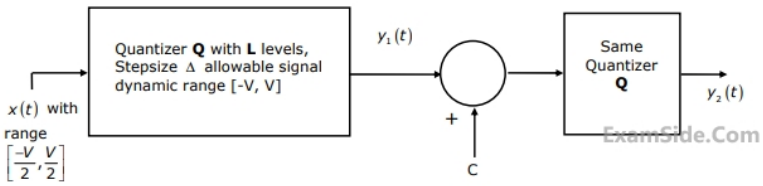
(D) The value of N0 will determine which of the two constellations has a lower probability of symbol error.

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Q32. GATE ECE 2006

Open solution

In the following figure the minimum value of the constant "C", which is to be added to $y_1(t)$ such that $y_1(t)$ and $y_2(t)$ are different, is



(A) Δ (B) $\frac{\Delta}{2}$ (C) $\frac{\Delta^2}{12}$ (D) $\frac{\Delta}{L}$

Q33. GATE ECE 2006

Open solution

Let $g(t) = p(t)^* p(t)$ where $*$ denotes convolution and $p(t) = u(t) - u(t-1)$ with $u(t)$ being the unit step function

The impulse response of filter matched to the signal $s(t) = g(t) - \delta(t-2)^* g(t)$ is given as:

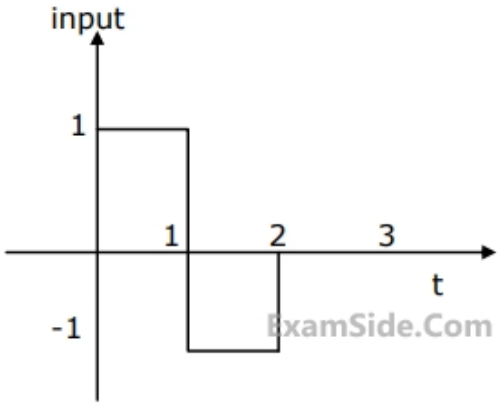
(A) $s(1-t)$ (B) $-s(1-t)$ (C) $-s(t)$ (D) $s(t)$

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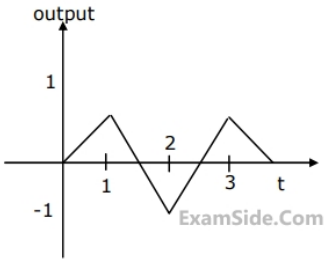
Q34. GATE ECE 2005

A signal as shown in figure is applied to a matched filter. Which of the following does represent the output of this matched filter?

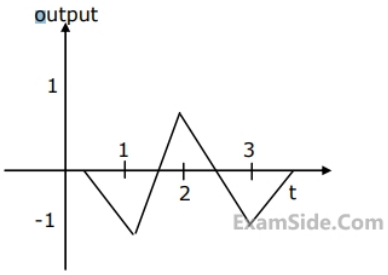
Open solution



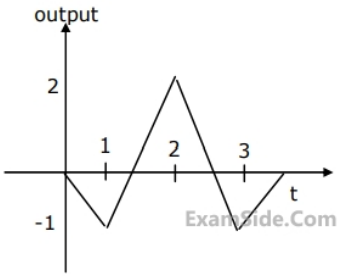
(A)



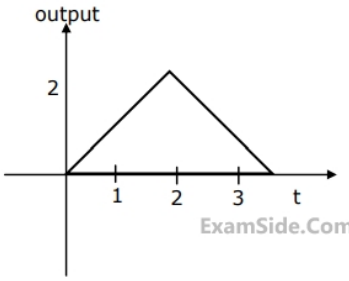
(B)



(C)



(D)

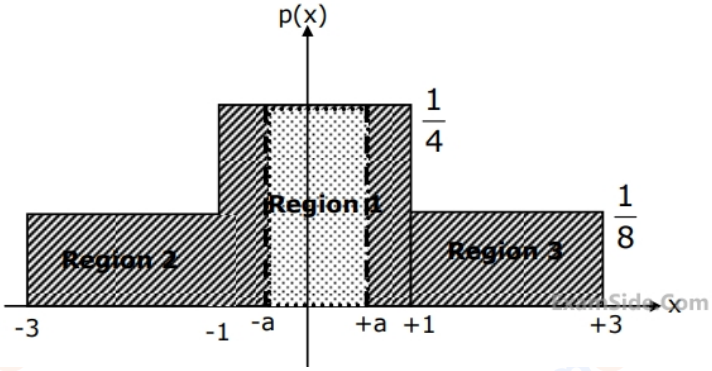


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V

Q35. GATE ECE 2005

A symmetric three-level midtread quantizer is to be designed assuming equiprobable occurrence of all quantization levels.
If the input probability density function is divided into three regions as shown in figure, the value of 'a' in the figure is



(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Open solution

Q36. GATE ECE 2005

A symmetric three-level midtread quantizer is to be designed assuming equiprobable occurrence of all quantization levels.
The quantization noise power for the quantization region between −a and +a in the figure is

(A) $\frac{4}{81}$ (B) $\frac{1}{9}$ (C) $\frac{5}{81}$ (D) $\frac{2}{81}$

Open solution

Q37. GATE ECE 2004

Consider a binary digital communication system with equally likely $0's$ and $1's$. When binary 0 is transmitted the voltage at the detector input can lie between the levels $-0.25V$ and $+0.25V$ with equal probability when binary 1 is transmitted, the voltage at the detector can have any value between $0$and $1V$ with equal probability. If the detector has a threshold of $2.0V$ (i.e., if the received signal is greater than $0.2V$, the bit is taken as 1), the average bit error probability is

(A) 0.15 (B) 0.2 (C) 0.05 (D) 0.5

Open solution

Q38. GATE ECE 2003

A sinusoidal signal with peak-to-peak amplitude of 1.536V is quantized into 128 levels using a mid-rise uniform quantizer. The quantization-noise power is

(A) 0.768 (B) $48 \times 10^{-6}V^2$ (C) $12 \times 10^{-6}V^2$ (D) $3.072V$

Open solution

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Q39. GATE ECE 2003

If E_b, the energy per bit of a binary digital signal, is 10^{-5} watt-sec and the one-sided power spectral density of the white noise, $N_0 = 10^{-6}$ W/Hz, then the output SNR of the matched filter is

(A) 26 dB (B) 10 dB (C) 20 dB (D) 13 dB

Open solution

Q40. GATE ECE 2001

During transmission over a communication channel, bit errors occur independently with probability 'p'. If a block of n bits is transmitted, the probability of at most one bit error is equal to

(A) $1-(1-p)^n$ (B) $p+(n-1)(1-p)$

(C) $np-(1-p)^{n-1}$ (D) $(1-p)^n+np(1-p)^{n-1}$

Open solution

Q41. GATE ECE 1999

The peak-to-peak input to an 8-bit PCM coder is 2 volts . The signal power - to -quantization noise power ratio (in dB) for an input of 0.5 $\cos(\omega_m t)$ is

(A) 47.8 (B) 49.8 (C) 95.6 (D) 99.6

Open solution

Q42. GATE ECE 1999

The input to a matched filter is given by

$$S(t)=\begin{cases}10\sin\left(2\pi\times10^6 t\right), & 0<1<10^{-4}\sec\\ 0 & , \quad otherwise\end{cases}$$

The peak amplitude of the filter output is

(A) 10 volts (B) 5 volts (C) 10 millivolts (D) 5 millivolts

Open solution

Q43. GATE ECE 1988

A signal having uniformly distributed amplitude in the interval (-V, +V) is to be encoded using PCM with uniform quantization. The signal - to - quantizing noise ratio is determined by the

(A) dynamic range of the signal (B) sampling rate

(C) number of quantizing levels (D) power spectrum of signal

Open solution

Q44. GATE ECE 1988

In a digital communication system, transmissions of successive bits through a noisy channel are assumed to be independent events with error probability p. The probability of at most one error in the transmission of an 8 - bit sequence is

(A) $7(1-p)/8+p/8$ (B) $(1-p)^8+8p(1-p)^7$

(C) $(1-p)^8+(1-p)^7$ (D) $(1-p)^8+p(1-p)^7$

Open solution

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Q45. GATE ECE 1987

Companding in PCM systems leads to improved signal - to - quantization noise ratio. This improvement is for

Open solution

(A) lower frequency components only

(B) higher frequency components only

(C) lower amplitudes only

(D) higher amplitudes only

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Answer Key

Marks 1

1. **7**	2. **C**	3. **32**	4. **0**	5. **C**

Marks 2

6. **B**	7. **D**	8. **0.040 to 0.041**	9. **0.028**	10. **A**
11. **A**	12. **0.4**	13. **D**	14. **C**	15. **C**
16. **C**	17. **B**	18. **B**	19. **D**	20. **D**
21. **D**	22. **B**	23. **D**	24. **C**	25. **C**
26. **A**	27. **C**	28. **A**	29. **D**	30. **B**
31. **A**	32. **B**	33. **C**	34. **C**	35. **B**
36. **A**	37. **A**	38. **C**	39. **D**	40. **D**
41. **B**	42. **D**	43. **C**	44. **B**	45. **C**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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